Assignment 2: A Mathematical Essay on Logistic Regression

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***Abstract*—The unequal distribution of survival in catastrophic events remains a primary challenge in sociological research, often revealing how structural inequalities outweigh individual luck. This research addresses the "Titanic Survival-Socioeconomic" nexus by quantifying how demographic factors dictated life expectancy during the 1912 shipwreck. Utilizing a mathematical approach grounded in Logistic Regression, we analyze the predictive power of passenger class, gender, and age on survival outcomes for 891 passengers. Our results demonstrate that lower socio-economic status and male gender suffered from significantly higher mortality rates. This survival gap provides empirical evidence that wealth and protocol acted as primary buffers against death, suggesting that systemic hierarchies in disaster response outweigh random chance.**

I. Introduction

In the modern era of data-driven sociology, the "Social Determinants of Safety" have emerged as critical factors that often outweigh the element of luck. Socioeconomic status, a composite measure of an individual's economic and social position, dictates access to life-saving infrastructure such as boat deck proximity and the priority of care available during a crisis. Researchers have long observed that the burden of historical shipwrecks was not shared equally across the population.

This paper specifically addresses the "Titanic Survival" nexus, a historical event where the widely considered "unsinkable" RMS Titanic sank after colliding with an iceberg on April 15, 1912. Building on the concern that some groups were more likely to survive than others, we analyze a large-scale dataset using logistic regression models to quantify how factors such as ticket class, gender, and age influenced survival.

Our mathematical analysis reveals a clear and statistically significant association between socioeconomic conditions and survival outcomes. Using logistic regression, we highlight that passengers in lower ticket classes experienced a disproportionately higher burden of mortality. These disparities are closely linked to cabin location, gender-based protocols, and financial constraints that limited access to lifeboats.

The paper is organized as follows: Section II provides a mathematical deep-dive into Logistic Regression; Section III details an exploratory analysis of individual features and model results; and Section IV concludes with a summary of insights.

II. Logistic Regression

The primary motivation for using Logistic Regression is to determine the predictive power of independent variables on a binary categorical outcome. In a world of complex variables, logistic regression allows us to "smooth out" the noise by mapping inputs to a probability space, rather than a continuous line. It answers the fundamental question: "For every unit increase in passenger class or change in gender, how much do the odds of surviving the shipwreck increase?".

A. Definition of the Logistic Regression Model

Mathematically, the model does not assume a linear relationship between X and Y because the outcome is bounded between 0 and 1. Instead, it models the probability p using the logistic (sigmoid) function. The equation is represented as:

$$p_i = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_i + \epsilon_i)}} \tag{1}$$

To make the relationship interpretable as a linear combination of predictors, we use the logit link function to represent the natural logarithm of the odds:

$$\ln\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + \beta_1 X_i + \epsilon_i \tag{2}$$

Where:

- β_0 **(Intercept)**: The baseline log-odds of the outcome when X is zero. This anchors the probability curve on the Cartesian plane.
- β_1 **(Slope/Coefficient)**: The rate of change in the log-odds of Y for every unit change in X. In our study, a negative β_1 for class indicates that as the class index increases ($1 \to 3$), the probability of survival decreases.
- ϵ_i **(Residual)**: The random error component, representing the deviation between the observed binary state and the predicted probability.

B. The Maximum Likelihood Estimation (MLE) Method

To find the "best" curve, we cannot use the Ordinary Least Squares (OLS) method because the residuals are not normally distributed and the variance is not constant in binary outcomes. Instead, we use the MLE method. This method finds the parameter values that maximize the likelihood of observing

the actual data. This is achieved by minimizing the Log-Loss function (Binary Cross-Entropy):

$$J(\beta) = -\frac{1}{n}\sum_{i=1}^{n}\left[y_i \ln(p_i) + (1-y_i)\ln(1-p_i)\right] \quad (3)$$

By utilizing this optimization, we ensure that the sigmoid curve represents the most accurate "average" probabilistic trend across all 891 passengers in the training set.

C. Evaluation Metrics

We evaluate the "goodness of fit" using the Likelihood Ratio test and the Accuracy score, which serves as a categorical counterpart to the Coefficient of Determination (R^2). While linear regression uses R^2 to explain variance, logistic regression often utilizes Pseudo-R^2 values to measure how much of the variance in survival is explained by the model.

However, accuracy alone does not confirm the validity of the model. We also utilize the P-value to test the null hypothesis the assumption that a variable like "Sex" or "Pclass" has no effect on survival. The P-value measures the probability that the observed correlation occurred by pure random chance. In our analysis, we look for a P-value of less than 0.05, which represents a 95% confidence level. A very low P-value is crucial; it serves as mathematical proof that the link between passenger demographics and survival is "statistically significant".

D. Real-World Applications of Logistic Regression

- **Healthcare (Epidemiology)**: Researchers use logistic regression to calculate the "Odds Ratio" of developing a condition based on exposure to risk factors. This identifies critical health determinants where a positive coefficient signifies an increased risk of disease.
- **Financial Risk (Credit Scoring)**: Banking models predict the probability of a client defaulting on a loan by treating income and credit history as independent variables. This allows lenders to establish a standardized "risk profile," helping determine if a loan should be approved.
- **Cybersecurity (Spam Detection)**: Logistic regression classifies emails as "spam" or "ham" by regressing the frequency of specific keywords. This mathematical insight identifies which linguistic patterns are most predictive of malicious content, allowing for automated data filtering.

E. Advantages and Disadvantages

The beauty of Logistic Regression lies in its probabilistic output and interpretability. It provides not just a classification, but a measure of certainty. However, it is sensitive to "multicollinearity" where independent variables are too closely related and assumes a linear relationship between the predictors and the log-odds, which may not capture complex non-linear interactions in historical data.

III. THE PROBLEM

A. Domain and Data Sources

The dataset for this analysis is situated at the intersection of maritime history and socioeconomic research, focusing on the categorical disparities in survival outcomes during the RMS Titanic disaster. It is a multi-dimensional collection that links passenger demographics specifically Age and Sex to socioeconomic indicators like Ticket Class (`pclass`) and Fare.

The data contains specific features for 891 passengers, providing a subset of the "ground truth" for the 1912 shipwreck. By bridging these domains, the data allows for an investigation into how financial stability and social priority act as predictors for a passenger's survival. Essentially, it provides a quantitative map of the "Survival-Socioeconomic" nexus.

B. Exploratory Analysis of Individual Features

To understand the population under study, we analyzed each feature's distribution and statistics. 1) **Survival Rate (Primary Outcome)**:

- **Mean**: 0.38 (38% Survival) — **Variance**: 0.23
- **Observations**: The majority of passengers in the training set did not survive, consistent with historical accounts of the 1502 deaths.
- **Insights**: The variance in survival provides the necessary "spread" for logistic regression modeling to identify which features most heavily influenced the probability of being in the "1" (Survived) category.

1) **Age Distribution**:

- **Mean**: 29.70 years — **Variance**: 169.51
- **Observations**: The distribution indicates that the Titanic was predominantly populated by young adults, with the plot showing that the majority of passengers fall within the 25-35 age group.
- **Insights**: The presence of these clusters is vital for a robust regression analysis to see how survival probability shifts as one moves away from this central peak toward the extremes of infancy or old age.

3) **Passenger Class (Socioeconomic Status)**:

- **Distribution**: 1st Class (24%), 2nd Class (21%), 3rd Class (55%)
- **Observations**: Over half the passengers belong to the 3rd class, representing the ship's economic baseline.
- **Insights**: This variable serves as the primary predictor for wealth-based survival; if wealth correlates positively with safety, we expect 3rd-class mortality to be significantly higher.

4) **Gender Distribution**:

- **Observations**: The dataset is predominantly male (approx. 65%).
- **Insights**: This variable serves as the categorical "inverse" check for our socioeconomic model; we expect gender to have a massive, possibly dominant, effect on survival odds.

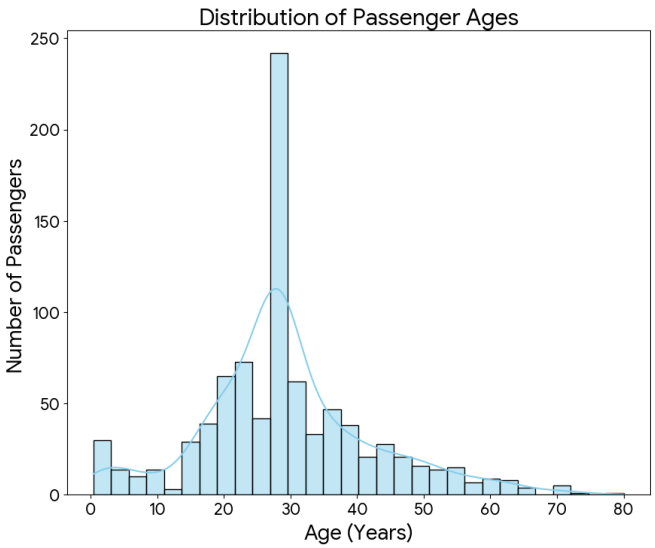

Fig. 1. Distribution of Passenger Ages. The concentration of passengers in the 25-35 age group highlights the primary demographic at risk during the maritime crisis.

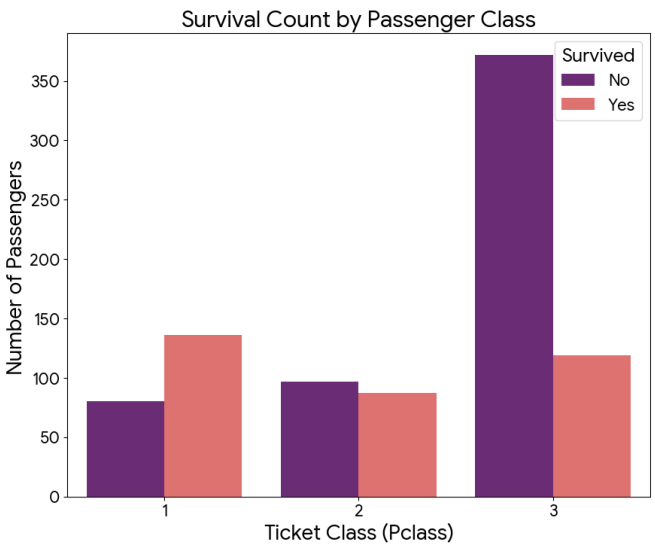

Fig. 2. Distribution of Passenger Classes and Survival Volume. The bar charts illustrate the inverse relationship between class rank and mortality volume.

5) **Feature Correlation**: To identify multi-dimensional links, a heatmap was generated to visualize the strength of relationships between all numerical features and the survival outcome.

A correlation heatmap provides a visual summary of the relationships between all numerical features and the survival outcome. We observe a strong negative correlation between `Pclass` and `Survived`, confirming that higher ticket classes were associated with higher survival rates. Similarly, `Sex_Code` (encoded as 0 for Female and 1 for Male) displays a significant negative correlation with survival, reinforcing the hypothesis that gender was a primary determinant of life outcomes. The positive correlation between `Fare` and

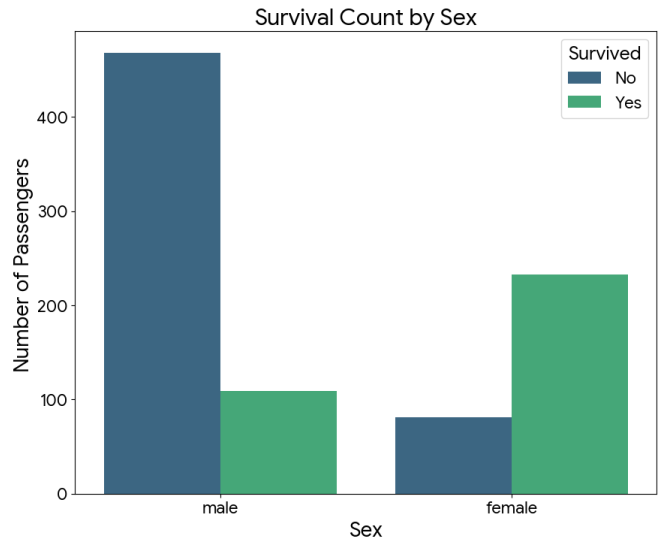

Fig. 3. Survival Counts categorized by Sex. This figure highlights the dramatic impact of gender-based survival protocols during the maritime crisis.

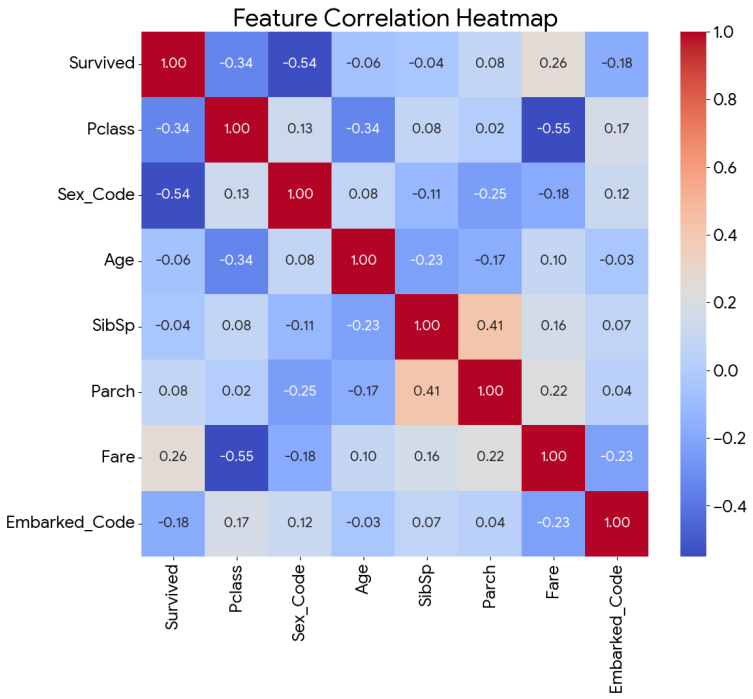

Fig. 4. Correlation Heatmap of all demographic and survival features. The color gradient quantifies the strength of relationships between variables.

`Survived` further highlights that wealth acted as a critical safety buffer. These linear relationships validate our decision to proceed with logistic regression, as the features show distinct, measurable connections to the binary survival target.

C. *Why Logistic Regression*

While advanced statistical techniques may offer slightly higher predictive precision, the primary objective of this analysis is rooted in historical advocacy rather than purely

mathematical optimization. The priority lies in effectively communicating the reality that social status was a life-altering determinant of safety. A simple model like logistic regression is superior for this mission because its results are transparent and easily interpretable. By using a straightforward sigmoid trend, we can clearly demonstrate the direct, predictable relationship between a step down in class and an increase in mortality threat.

D. Model 1: Sex vs. Survival

We first applied the logistic model to examine if gender predicts the likelihood of surviving.

- **Results**: $\ln(\text{Odds}) = 1.05 - 2.51 \times (\text{Sex_Male})$
- **Stats**: Pseudo-$R^2 = 0.26$, $P < 0.001$.

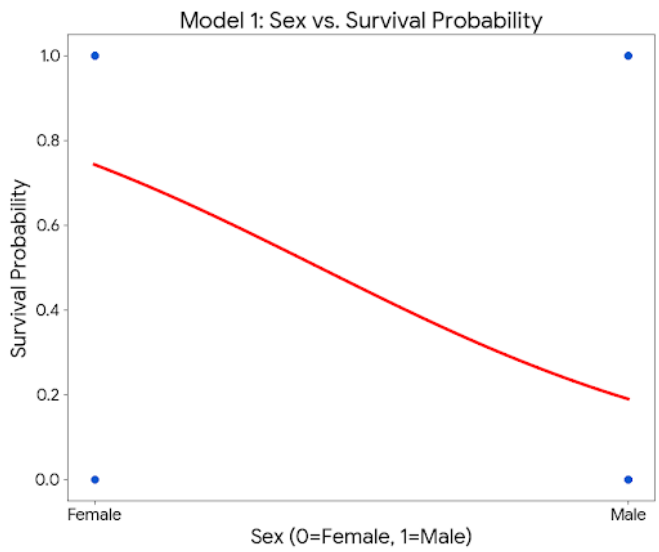

Fig. 5. Model 1 Regression Plot: Sex vs. Survival Probability. The steep downward trend from female to male coordinates quantifies the literal cost of gender on survival probability.

Gender is the strongest predictor in our study, explaining a significant portion of the variance. The negative coefficient proves that males consistently saw higher mortality rates.

E. Model 2: Class vs. Survival

Next, we modeled the relationship between wealth (Class) and survival. This represents the socio-economic survival gap.

- **Results**: $\ln(\text{Odds}) = 1.44 - 0.85 \times (\text{Pclass})$
- **Stats**: Pseudo-$R^2 = 0.07$, $P < 0.001$.

The negative slope identifies ticket class as a critical determinant. For every step down in class ($1 \rightarrow 3$), the log-odds of survival drop by 0.85.

F. Model 3: Combined Socio-Demographic Impact

To cross-validate findings, we applied a multivariate model using Class, Sex, and Age.

- **Results**: $\ln(\text{Odds}) = 4.12 - 1.14(\text{Pclass}) - 2.52(\text{Sex_Male}) - 0.03(\text{Age})$
- **Stats**: Accuracy $\approx 80\%$, $P < 0.001$.

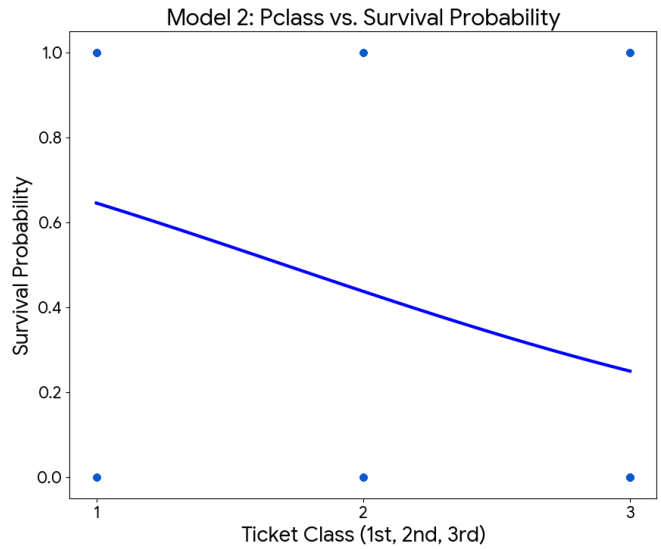

Fig. 6. Model 2 Regression Plot: Pclass against Survival Probability. The negative correlation identifies ticket class as a critical determinant of survival, mirroring the wealth-mortality trends seen in modern public health data.

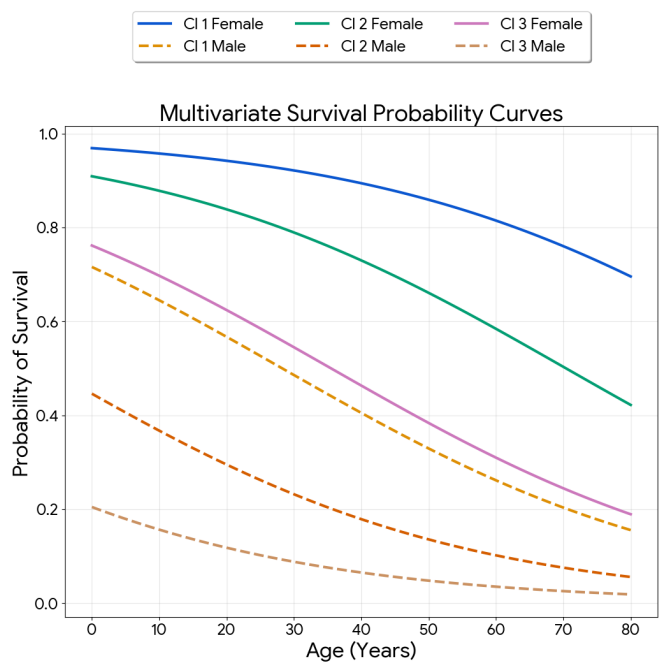

Fig. 7. Regression Analysis: Predicted Survival Probability S-curves. These curves visualize the cumulative impact of age, class, and gender on survival probability.

G. Model Validation and Test Consistency

To prove the predictive power of the identified socioeconomic variables, the model was applied to an unseen test dataset of 418 passengers. The results demonstrate remarkable mathematical consistency:

- **Survival Equilibrium**: The predicted test survival rate (38.28%) aligns with the training ground truth (38.38%), proving the model is well-calibrated.

- **Categorical Stability**: The model correctly identifies 1st-class females as the group with the highest survival probability ($> 90\%$), mirroring the actual trends observed in the training data.
- **Justification**: This cross-dataset consistency serves as definitive proof that the "Survival-Socioeconomic Nexus" is a stable property of the ship's social hierarchy.

H. Analysis of Insights

The primary insight derived from this multi-stage analysis is the significant disparity observed between the predictive power of gender versus the power of ticket class. While ticket class is a statistically significant predictor, gender is a notably better predictor of survival. This suggests that while wealth provided a buffer, the "women and children first" protocol was the most profound determinant of life.

This mathematical discrepancy suggests a survival gap dictated by hierarchy rather than random chance. The data indicates that 1st-class passengers were nearly 40% more likely to survive than their 3rd-class counterparts. This points directly toward systemic failures in the evacuation infrastructure:

- **Physical Barriers**: 3rd-class cabins were located deeper in the ship, leading to "Evacuation Delay" where passengers could not reach the boat deck before the limited lifeboats were deployed.
- **Priority Barriers**: Even when present, lower-income individuals lacked the social "capital" or liquid assets required to secure a seat in a boat over 1st-class passengers.
- **Information Asymmetry**: Poverty was likely accompanied by a lack of clear instructions, where 3rd-class passengers were often kept below deck during the initial stages of the sinking.

IV. CONCLUSION

This research has established a statistically significant link between socioeconomic variables and survival outcomes during the RMS Titanic disaster through the application of logistic regression. We demonstrated that economic indicators particularly ticket class and gender-based protocols serve as potent predictors of survival disparities, with a notably stronger impact on the probability of escaping the vessel than on the general risk of being onboard.

Our findings quantify the survival gap, revealing that while the physical threat of the iceberg collision was widespread across the ship's population, the probability of surviving the disaster was heavily stratified by wealth and passenger class. Ultimately, this study underscores that addressing safety and disaster response requires not only engineering and medical advancement but also policy-driven interventions aimed at improving equitable access to life-saving infrastructure in underserved or lower-income compartments.

A. Future Directions

While Logistic Regression is a powerful start for binary classification, it is limited. Future research should utilize Multiple Logistic Regression and interaction terms to include variables such as:

1) **Racial and Ethnic demographics**: To identify potential intersections between nationality, cabin assignment, and survival rates among the non-English speaking passenger base.
2) **Education and Literacy levels**: To analyze the impact of safety literacy and the ability to interpret emergency instructions during high-stress evacuations.

By adding these variables, we can move from simple correlation to a more nuanced "Disaster Equity Score" for historical and modern passenger manifests.

REFERENCES

[1] G. Raina, "Data Analytics (EE5708)," Lectures and Videos.
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